

Controlling the Dynamics of Cardiac Muscle Using Small Electrical Stimuli

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1. Introduction

Understanding how systems evolve from one moment to the next and developing quantitative mathematical models to predict an observed dynamical behavior is an active area of research, especially when the systems are nonlinear. In a nonlinear system, the evolution of one variable depends in a nonlinear fashion on the collection of system variables. It is found that nonlinear dynamical systems can display a variety of exceedingly complex behaviors in comparison to their linear counterparts. For example, a nonlinear system can undergo a bifurcation, where one type of dynamical behavior (e.g., a steady temporal evolution of the variables) switches over to a different type of behavior (e.g., a periodic oscillation of the variables) as a system parameter is varied. Also, researchers have found that seemingly simple nonlinear systems characterized by as few as three dynamical variables can display a behavior known as chaos. In a chaotic system, the variables fluctuate in a seemingly erratic, 'noisy' fashion and they display a property known as extreme sensitivity to initial conditions where the slightest perturbation to the system completely changes its long-term state.

Research on nonlinear dynamical systems displaying temporal instabilities and chaos has progressed at a truly astounding pace. It is now possible to control chaos, delay the onset of bifurcations and instabilities, and direct a trajectory to a desired target state by applying only minute perturbations to the system. The control strategies are quite general and do not require a detailed mathematical model of the system. The key idea is to design perturbations that direct the system near to or stabilize it about one of the unstable periodic orbits embedded in the system [1,2]. Motivated by successes in applying these methods to a variety of physical systems, recent attention has focused on controlling biological systems such as the heart [3]. For example, Hall et al. [4] used closed-loop feedback to suppress a period-doubling bifurcation in atrio-ventricular nodal conduction of an *in vitro* rabbit heart model, which is designed to mimic a human cardiac arrhythmia.

Another potential application of closed-loop feedback methods is to terminate or prevent the occurrence of fibrillation by applying small electrical stimuli to the heart muscle. Fibrillation, often the cause of sudden cardiac death, is a state in which the average heart rate is elevated and the muscle contractions are spatially uncoordinated, preventing the heart from pumping blood effectively. The most common strategy for terminating fibrillation is to apply a large electric shock designed to stop temporarily all cardiac activity, with the hope that a normal rhythm will start with the next heart beat [5–7]. Such a strategy alters significantly the dynamical state of the heart, is not always successful, and is often quite painful to the patient.

A more subtle approach to this problem is to administer occasional, small electrical stimuli to the heart muscle, where the strength and timing of the shocks are

based on a nonlinear dynamics control algorithm. The potential success of this approach relies on the understanding that the fibrillating heart is a complex, yet deterministic, nonlinear dynamical system [8].

Several hurdles have to be overcome before this approach can be successful. One primary issue is that the heart displays complex behavior as a function of time and space [9,10], which is believed to be a manifestation of the deterministic behavior known as *spatio-temporal chaos* [11]. It is not clear whether the chaos control algorithms, designed originally for suppressing *temporal* instabilities, can be applied to fibrillation. Controlling spatio-temporal complexity using small perturbations is interesting from a fundamental nonlinear dynamics perspective because it is not yet firmly established whether a generic system can be characterized by unstable period orbits, and no general control algorithms have been established. One suggested approach is to place standard chaos controllers designed to suppress temporal instabilities at a few or many spatial locations [12]. A recent promising study suggests that it may be possible to stabilize *in vivo* human atrial fibrillation by applying control at a single spatial location [13].

In a systematic approach to controlling heart muscle dynamics, we have used an *in vitro* animal model consisting of a small piece of periodically paced bullfrog (*Rana catesbeiana*) cardiac muscle to test nonlinear dynamics control methods in a situation where spatial complexity is not an issue. Ultimately, we will explore their application to the whole fibrillating heart. Our experimental test-bed can also be used to investigate other local dynamical behaviors, such as bistability and hysteresis, which may also play a role in the initiation of fibrillation [14].

In this chapter, we present a survey of current progress in the control of cardiac instabilities and discuss some of our recent results. We first discuss general aspects of the heart as a dynamical system (Section 2), the idea of closed-loop feedback control in complex nonlinear systems (Section 3), and its application to cardiac dynamics (Section 4). We review some of our recent results on spatially localized dynamics in bullfrog cardiac muscle (Section 5), and the transitions of this system between bistable states using small perturbations (Section 6). Finally, we will discuss the application of chaos control techniques to this system (Section 7) and the implications of these results for applications of control in the whole heart (Section 8).

2. The heart as a dynamical system

The heart is a complex nonlinear system, designed to pump blood effectively. Its mechanical contractions are mediated by waves of electrochemical excitation that sweep through specialized conduction systems, such as the atrio-ventricular and sino-atrial nodes, Purkinje fibers, and through the heart muscle. These waves of excitation are followed by a refractory period when the tissue cannot respond to further stimulation, a fundamental characteristic of the generic, idealized dynamical system classified as an ‘excitable media’.

In healthy heart conditions, the waves of excitation cause a coordinated contraction of the muscle starting near the top of the right atrium at the sinoatrial

node, progressing to the ventricles through the atrioventricular node and the Purkinje fibers, and terminating near the base of the heart. This type of behavior is known as normal sinus rhythm. In some situations, even in a nominally healthy heart, this orderly procession of waves can evolve into a complex dynamical state known as fibrillation in which the muscle contractions are uncoordinated [8]. Fibrillation in the atria is usually not life threatening, but can lead to an increased risk to stroke or ventricular fibrillation and often leaves a person feeling tired or having a lack of energy. Ventricular fibrillation, however, results in a rapid loss of blood pressure, leading to death within a few minutes if left untreated.

To gain an understanding of the dynamical origins of this behavior, we undertake a brief review of cardiac electrophysiology covering the cellular-level dynamics as well as the macroscopic behavior of the whole heart.

2.1. Cellular and small-tissue dynamics

A heartbeat is initiated when specialized heart muscle cells in the sino-atrial node generate an electrical signal, known as the action potential, which then propagates through the rest of the heart by myocardial cells specialized for conduction of the electrical signal. As the wave of activation propagates through the tissue, myocardial fibers contract in response to changes in the intracellular calcium concentration initiated by the action potential [15,16].

A schematic diagram of a cardiac cell plasma membrane is shown in Fig. 1. The intracellular and extracellular solutions contain different concentrations of ions, primarily potassium (K^+), sodium (Na^+), chloride (Cl^-), and to a lesser extent calcium (Ca^{2+}). The interior and exterior regions are separated by a lipid bilayer approximately 3 nm thick that is impermeable to the ions. Proteins in the membrane selectively carry ions across the membrane under certain conditions of voltage and concentration. The concentration gradient of each species of ion across the cell membrane causes a potential difference across the membrane, which drives the

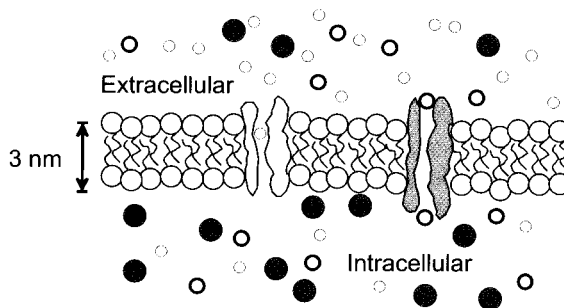


Fig. 1. An illustration of a plasma membrane. The membrane consists of a lipid bilayer that keeps the solution within the cellular structure separate from the solution surrounding the tissue. Ions are transported across the membrane wall by proteins embedded in the membrane that act as ion-selective, voltage-sensitive channels and pumps.

cellular activity. In the cell's non-activated or resting state, this difference is approximately -90 mV.

The response of the cardiac tissue to electrical stimulation is a result of the voltage-dependent conductances of the protein channels that govern ion transfer across the membrane. In turn, the voltage across the membrane depends on the interaction between the nonlinear conductances of the proteins and the transmembrane voltage. If the voltage difference across the membrane increases above a critical level, some of the sodium channel proteins open leading to a slight depolarization of the cell (i.e., an increase of the transmembrane potential), which in turn opens more channels. This leads to a rapid depolarization of the membrane (upstroke of the action potential), resulting from the inflow of sodium ions through the open channels.

The primary reason that the heart is capable of beating periodically arises from the nonlinear behavior of the membrane dynamics. The nonlinearity arises from the threshold behavior of the sodium channel proteins: the proteins are only activated once the small potential differences across the membrane exceed a critical value. While such nonlinearity is necessary for normal heart function, it also is responsible for the complex behaviors displayed by the heart such as fibrillation.

The cardiac cell depolarization ultimately activates potential-dependent potassium channels, and an efflux of potassium follows the sodium influx. The potassium efflux results in the action potential downstroke. In cardiac cells (but not in neurons) the action potential is prolonged by an influx of calcium, which is essential for the contraction of the myocardial fibers. During this plateau, known as the refractory period, the cell is insensitive to further electrical stimulation. On a longer time scales, sodium (potassium) is redistributed to the outside (inside) of the cell by another membrane protein, the $\text{Na}^+/\text{K}^+/\text{ATPase}$, which uses energy from the hydrolysis of the ATP molecule to pump the ions back to their respective sides of the membrane.

The transmembrane voltage can be measured by inserting a microelectrode through the plasma membrane using a setup shown schematically in Fig. 2. Using this setup, we measure the *in vitro* temporal evolution of the transmembrane voltage of a small piece of periodically paced bullfrog (*Rana catesbeiana*) ventricular myocardium as shown in Fig. 3a. The trace at the bottom of the panel shows the stimulus waveform. In the bullfrog myocardium, the action potential duration is typically between 300 and 900 ms; its height is about 90 mV.

In some experiments on larger pieces of cardiac tissue, it is difficult to maintain the insertion of many microelectrodes in cardiac cells when attempting to record the electrical activity of the heart at many spatial locations because of the mechanical motion of the heart. One recently discovered method for making such measurements involves the use of voltage-sensitive dyes that are embedded in the cell membrane. When the dye-impregnated tissue is illuminated with light, the amount of fluorescence emitted by the dye is linearly proportional to the transmembrane voltage. Ultra-high speed and sensitivity cameras then allow the direct visualization of the transmembrane voltage [9,10]. However, the signals are usually small and the dyes can be toxic to the tissue.

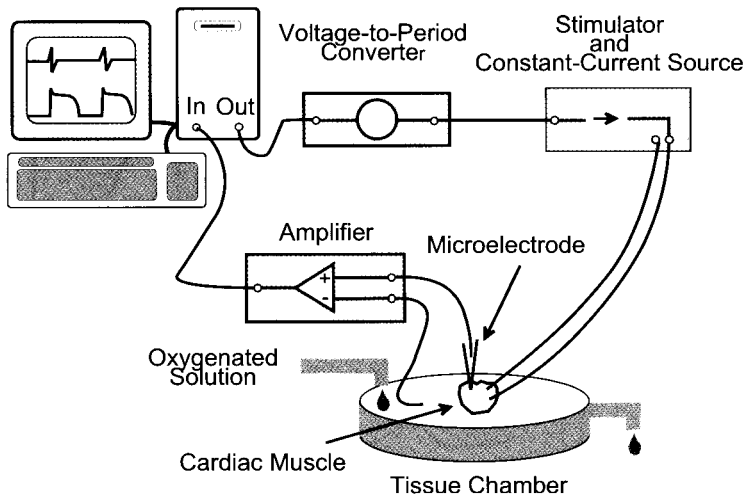


Fig. 2. Experimental setup to investigate *in vitro* the dynamics of rapidly paced cardiac muscle consisting of three subsystems. The tissue is bathed with flowing physiological solution to keep it alive. The computer system controls the generation of brief, constant current periodic stimuli that are applied to the tissue through two fine tungsten wires. The transmembrane potential is measured with a microelectrode, amplified, and digitized using a computer interface.

A more common approach to measuring cardiac dynamics is to use extracellular electrodes. In this technique, a fine wire is placed on the surface of the tissue and the potential difference between this electrode and another placed at a convenient reference location is measured. The extracellular signal is related to the current flowing from the cells located near the electrode and is roughly proportional to the second derivative of the transmembrane voltage. An extracellular recording from a bullfrog is shown in Fig. 3b and is compared with the transmembrane voltage recorded simultaneously using a microelectrode. While the morphology of two signals is quite different, the beginning and end of the action potential are clear from the extracellular signal. Unfortunately, the deflection in the extracellular signal signifying the repolarization is not always detectable.

The electrodynamics of the cell membrane, such as that shown in Fig. 3, can be modeled by an equivalent electric circuit shown schematically in Fig. 4. The flow of each ion across the membrane can be treated as a current through a series combination of a battery and a variable conductance. The battery represents the voltage generated by the equilibrium concentration gradient of a particular ion. The variable conductances represent the voltage-dependent ion-selective proteins in the membrane that move ions across the membrane. The additional current through the resistor and capacitor represents a voltage-independent ion flow across the membrane and the membrane capacitance.

Modeling the particular ion flows and the voltage across a membrane including the nonlinear conductances involves a large number of differential equations whose

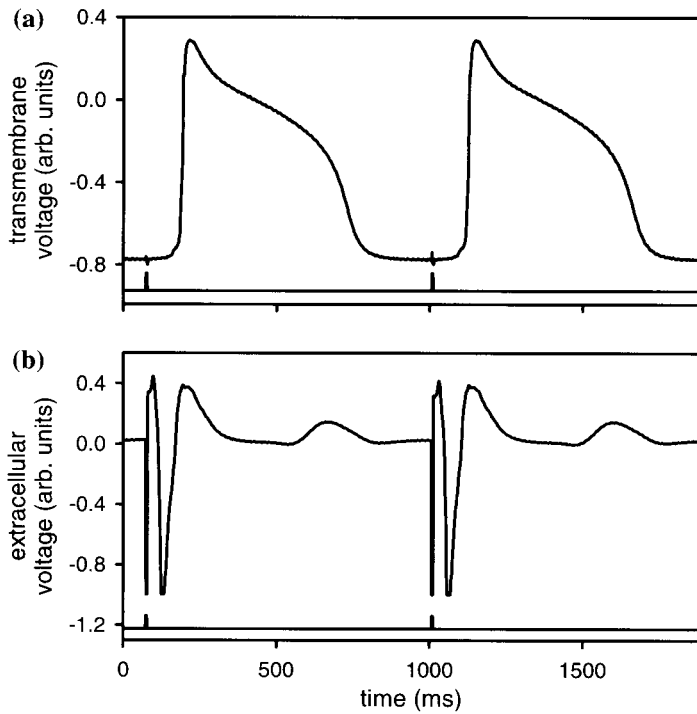


Fig. 3. Temporal evolution of the (a) transmembrane potential and (b) extracellular potential recorded from a small piece of bullfrog cardiac tissue that is stimulated periodically. The transient depolarization of the membrane is called an action potential. The stimulus waveform is shown at the bottom of the panel.

precise details depend on the cell type, the animal species [17] and the specific type of tissue [18,19]. Nevertheless, all cardiac muscles possess some of the general characteristics of excitable media, and simplified models of the tissue dynamics have been developed that give a reasonable description of the observed behaviors [20,21]. Experimentally, the duration of the action potential (denoted by APD) is seen to depend approximately on the time interval between the end of one action potential

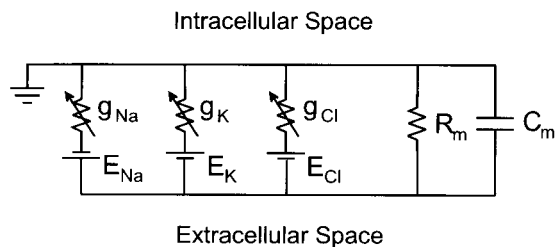


Fig. 4. An equivalent circuit model of a cardiac cell.

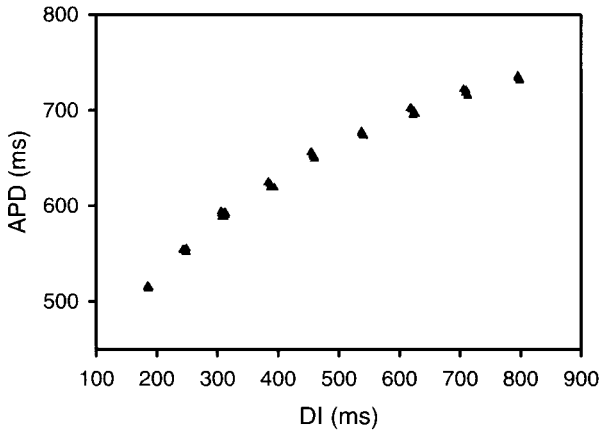


Fig. 5. Experimentally observed restitution relation of an *in vitro* periodically paced small piece of bullfrog cardiac muscle.

to the beginning to the next, called the diastolic interval (denoted by DI). The APD is assumed to be a nonlinear function f of the DI in a form given by

$$APD_n = f(DI_n). \quad (1)$$

The relationship between APD and the preceding DI is called a restitution relation, shown in Fig. 5 for a small piece of periodically paced bullfrog cardiac tissue. Notice the slope of the restitution relation begins to level off for long DI, and there is a maximum APD. From an electrophysiology point of view, this relation makes sense because of the fact that the ionic pumps take some time to restore the equilibrium ionic balance. If the tissue is stimulated before this balance is restored, the action potential is shortened. This description clearly discards much information about the specific nature of the tissue, but is useful to understand the gross features of cardiac dynamics [21,22].

2.2. Whole-heart dynamics

While our understanding of single-cell dynamics has developed to a high level over the past decades, our understanding of the behavior of the whole heart is still a work in progress both from an experimental as well as the theoretical perspective. Whole-heart dynamics are more complicated because the heart consists of many million interconnected cells. Each cell is coupled to its neighbors through gap junctions, or channel proteins at the end of the cell. These channel proteins connect to the channel proteins in the neighboring cell, providing a path for current to flow from one cell to the next. This path couples the ionic concentration of a cell through diffusion to the cells around it, and allows for the coordinated spread of the action potential through the tissue. The ionic coupling between cells is responsible for the propagation of waves of excitation. It is interesting to note that the refractory period of individual cells results in the unidirectional nature of wave propagation in the heart. Cells in

front of a wave of excitation have not yet been activated, while those behind have recently been excited and consequently are refractory. This also explains why waves annihilate each other when they collide, instead of passing through each other. A detailed review of the mathematics of the propagation action potential through the heart is presented, for example, by Henriquez [23].

To gain some understanding of the spatiotemporal aspects of signal propagation in the heart, we have measured the electrical activity of a sheep right atrium *in vivo* during normal sinus rhythm using a multi-channel electronic mapping system [24–27] in collaboration with researchers in Duke University's Experimental Electrophysiology Laboratory. In this experiment¹, a dense array of 324 fine wires embedded in a flexible substrate is affixed to the epicardium of the right atrial chamber heart during open heart surgery. Extracellular electrograms are recorded from the tip of each electrode and processed to detect the occurrence of a muscle contraction beneath each electrode.

The temporal pattern of activation during normal sinus rhythm is shown in the series of panels in Fig. 6. Each panel corresponds to a 4.5 ms time-slice recorded during a single experiment. The gray and black dots in each panel represent the spatial location of the electrodes. A black dot indicates the detection of an activation during the time slice. Arrows have been added to guide the eye through the time course of the wave. The wave of excitation originates in the sinus node (off the plaque in the upper left region) and propagates across the atria. The wave of excitation shown in this figure corresponds to the 'P-wave' part of cardiac cycle and repeats approximately once a second. The total propagation time across the plaque (33 mm width) is about 31 ms.

This coordinated behavior can give way to complex, fast rhythm known as fibrillation (even in a nominally healthy heart) for reasons that are not entirely well understood. In the laboratory, we can induce fibrillation by giving a short burst of small electrical shocks to the tissue via one of the plaque electrodes. Fig. 7 shows a series of panels recorded from the same sheep but during atrial fibrillation. These panels are similar to the ones shown in Fig. 6, except that each panel corresponds to a 9.5 ms time slice of a single time series because the waves propagate more slowly during fibrillation. Note that the various portions of the muscle are constantly contracting during fibrillation, which contrasts with normal sinus rhythm where a

1 All procedures are approved by the Duke University Institutional Animal Care and Use Committee (IACUC) and conform to the Research Animal Use Guidelines of the American Heart Association. Female 61 ± 2 kg sheep are initially anesthetized with ketamine hydrochloride (15–22 mg/kg IM). Once anesthesia is achieved, the animal is intubated with a cuffed endotracheal tube and ventilated with a North American Drager model SAV ventilator. Isoflurane gas (1–5%) is administered continuously to maintain adequate anesthesia. A nasogastric tube is passed into the stomach to prevent rumen aspiration. Femoral arterial blood pressure and the lead II electrocardiogram are continuously displayed and monitored. Blood is withdrawn every 30–60 min to determine pH, P_{O_2} , P_{CO_2} , total CO_2 , based excess and the concentrations of Ca^{2+} , K^+ , Na^+ , and HCO_3^- . Normal physiological levels of the above are maintained by adjusting the ventilator and by IV injection of electrolytes. The chest is opened through a median sternotomy and the heart is suspended in a pericardial cradle. A 7.5×3.5 cm² mapping plaque is affixed to the epicardium of the right atrial wall with sutures to the pericardial cradle and the fatty tissue in the atrioventricular (AV) groove.

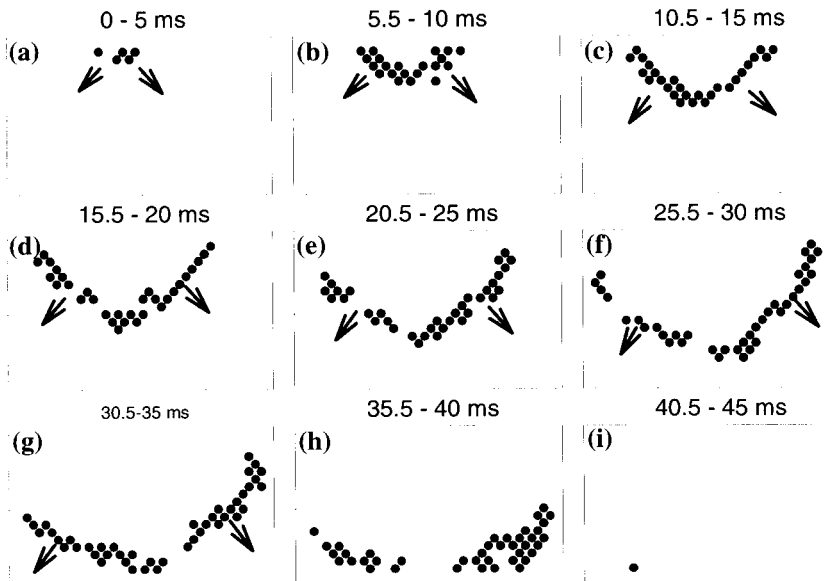


Fig. 6. Spatio-temporal evolution of a normal heart beat recorded *in vivo* from a sheep. The wave originates in the sinus node (off the electrode array near the top middle of the array). Each panel shows the location of a detected activation (black dot) during a 6 ms time-slice and the wave takes approximately 31 ms to travel across the electrode array. The array is approximately 7.5×3.5 cm and is affixed to the right atrium. Similar activity is observed to occur approximately once every second.

contraction occurs about once per second. During atrial fibrillation, there is no single point of origin for the excitation, the observed activity is self-sustaining, the shape of the wave is uneven, and its speed varies widely from location to location.

It is now accepted by most researchers that the uncoordinated, seemingly random muscle contractions characteristic of ventricular or atrial fibrillation possess some degree of order analogous to the spatio-temporal complexity displayed in physical systems, such as certain chemical reactions or wide-aperture optical systems. While this important discovery has changed our view of cardiac dynamics significantly, a detailed quantitative understanding of the mechanisms that trigger the heart to enter an episode of fibrillation, or the dynamical techniques necessary to control such an episode, remain elusive. A primary question is whether nonlinear dynamics control methods can prevent the occurrence or terminate this behavior. While nonlinear dynamics techniques have been used to successfully modulate local dynamics in cardiac tissue [3,4,28,29], it remains an open question whether such control techniques can be extended successfully to the spatiotemporal case of the *in vivo* beating heart, or whether nonlinear dynamics control methods can successfully stabilize cardiac rhythms. Other unresolved questions include the existence of precursors indicating that an episode of fibrillation is imminent and how large shocks applied to the heart act to terminate fibrillation.

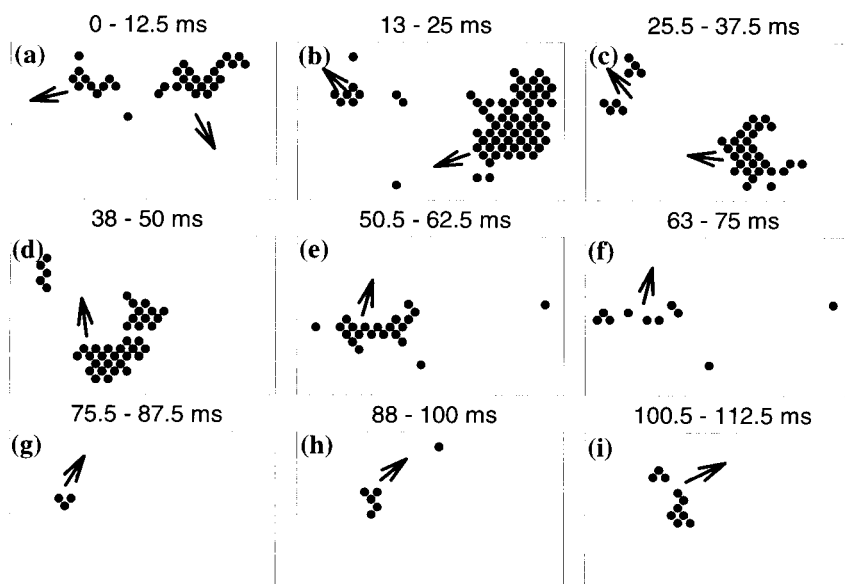


Fig. 7. Spatio-temporal evolution of electrochemical activation in an *in vivo* fibrillating sheep right atrium. Each panel represents a 12 ms time-slice. The wave of excitation is continuous and takes approximately 120 ms to complete one cycle around the atrium.

3. Controlling chaos

Feedback schemes for controlling chaos using small perturbations to an accessible system parameter, first proposed by Ott, Grebogi and Yorke (OGY) [30], are quite general, based on real-time measurement of the system dynamics, and do not require a detailed model of the dynamics. The key idea underlying the control schemes is to take advantage of the unstable dynamical states embedded within the system. Rather than making large changes to the system, the feedback technique applies small perturbations designed to stabilize one of these unstable states. As the system approaches the desired state, the strength of the perturbations required to keep it there vanishes, so that the feedback signal can be as small as the noise level in the system.

In the simplest implementation of the OGY idea, a dynamical variable characterizing the state of the system is measured at discrete time intervals, the difference from the measured and desired value (known *a priori*) is determined, and a control parameter is adjusted accordingly. For example, the dynamics of an electrical circuit can be controlled using a measured voltage as the dynamical variable and injecting current into a node of the circuit as the accessible system parameter (a procedure used in [31]). Variations of this method have been applied successfully to stabilize the dynamics of mechanical, electrical, optical, and chemical systems [1,2].

To illustrate the use of feedback control methods in the context of cardiac dynamics, we consider the seemingly simple situation in which a small piece of cardiac tissue is paced periodically by an external stimulus. As described earlier, the

dynamics of the tissue is characterized approximately in terms of the APD and DI through the nonlinear relation (1). For periodically paced tissue, there is an additional constraint between the dynamical properties of the tissue and the applied stimulus, leading to the following relation

$$DI_n = N * PP - APD_n, \quad (2)$$

where PP is the pacing period, N the smallest integer satisfying the condition $N * PP - APD_n > DI_{\min}$, and $DI_{\min}$ is the minimum diastolic interval the tissue can sustain [20,21]. Combining Eqs. (1) and (2) results in a mathematical mapping relating the future APD in terms of its past value in a form given by

$$APD_{n+1} = f(N * PP - APD_n). \quad (3)$$

For some tissue preparations, including bullfrog myocardium [32], a restitution relation of the form

$$f(DI_n) = APD_{\max} - A e^{-DI_n/\tau}, \quad (4)$$

gives good agreement with experimental observations, where $APD_{\max}$, A , and τ are tissue-dependent constants.

For long PP , the behavior of the paced muscle settles down to a regular periodic behavior after an initial transient. In this case, $DI \gg DI_{\min}$ and every stimulus produces an action potential whose shape and duration are identical so that $N = 1$ (this is often referred to as 1:1 or period-1 behavior). The steady-state action potential duration, known as the fixed point of the map, is denoted by APD^* and is given by the solution of the transcendental relation

$$APD^* = f(N * PP - APD^*). \quad (5)$$

It can also be determined graphically by finding the intersection of the restitution (4) and pacing (2) relations, as shown in Fig. 8. Note that the pacing relation line shown in the figure shifts to the left as PP decreases so that APD^* decreases as the pacing becomes faster.

The stability of the fixed point is governed by the value of $DI_{\min}$ and the slope of the restitution relation evaluated at the fixed point

$$\mu = \left. \frac{\partial f}{\partial APD} \right|_{APD=APD^*} = -\frac{A}{\tau} e^{-(PP-APD^*)}, \quad (6)$$

where μ is known as the Floquet multiplier. When the actual value of the APD is close to the fixed point, the temporal evolution of the action potential duration is given approximately by

$$\delta APD_{n+1} = \mu \delta APD_n, \quad (7)$$

where $\delta APD_n = APD_n - APD^*$. By inspection of Eq. (7), any slight perturbation that moves the system away from the fixed point will change by a factor $|\mu|$ on each iterate of the map. Therefore, the fixed point is unstable (the perturbation grows)

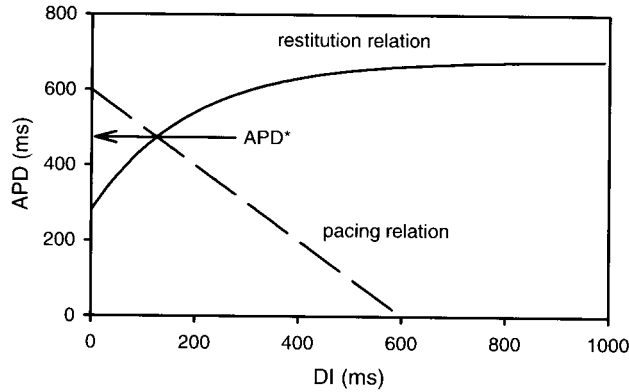


Fig. 8. A mathematical model of the restitution relation (solid line), and the pacing relation (dashed line) describing the constraint between PP, APD, and DI. The intersection of the two curves gives APD^* .

whenever $|\mu| > 1$. In addition, the fixed point will become unstable whenever $PP - APD^* < DI_{\min}$.

As PP decreases (thereby decreasing APD^* and increasing $|\mu|$), the fixed point becomes unstable and the tissue response switches over to a new pattern whose form depends on the precise values of the model parameters. Such a transition in the response pattern as a system parameter is swept (PP in this case) is known as a bifurcation and is one typical characteristic a nonlinear dynamical system.

For tissue where $DI_{\min}$ is large and characterized by a shallow restitution relation, the 1:1 behavior first becomes unstable when $PP - APD^* < DI_{\min}$ and is replaced by a pattern in which an action potential is induced for every other stimulus ($N = 2$). That is, the pacing is so rapid that the stimulus occurs during the period when the tissue is refractory and hence cannot respond with a new action potential. An action potential is only induced on the next stimulus after the tissue has recovered. This is known as a subcritical bifurcation and the response pattern is often referred to as 2:1. Experimental observations of this behavior are described in Section 5.

In a situation where $DI_{\min}$ is small ($PP - APD^* > DI_{\min}$ for all PP considered) and the tissue is characterized by a steep restitution relation, the 1:1 response pattern becomes unstable when $\mu < -1$ and is replaced by a pattern in which an action potential is induced by every stimulus but the duration alternates between two different values. This pattern is referred to as alternans, period-2 behavior, or 2:2 behavior by various authors. Alternans is sometimes observed in human electrocardiograms recorded from electrodes placed on the body surface, and can be correlated with an increased risk of a future episode of sudden cardiac death. Hence, it is important to understand the mechanisms giving rise to alternans and methods for controlling or suppressing this dynamical state.

The transition to alternans as PP is varied and can be visualized using a bifurcation diagram, as shown in Fig. 9, for tissue parameters $APD_{\max} = 680$ ms, $A =$

400 ms, $\tau = 190$ ms, and $DI_{\min} = 80$ ms. To generate the diagram, map (3) is iterated to determine its asymptotic behavior for each value of PP and the resulting APD for several iterates of the map are plotted. The value of PP is then changed and the procedure repeated. For long PP, only one value of APD appears, indicating a stable 1:1 pattern. For decreasing PP, the APD shortens, consistent with normal cardiac tissue restitution properties. At $PP = 631$ ms, an abrupt transition from a 1:1 to a 2:2 pattern occurs where it is seen that the APD alternates between short and long values and $\mu < -1$. This type of transition is called a forward or supercritical bifurcation [33].

Even though the 1:1 pattern is unstable for $PP < 631$ ms, there still exists a solution to Eq. (5) for APD^* . The value of this unstable fixed point as a function of PP is shown as the dashed line in Fig. 9. A properly designed controller can stabilize this state (the 1:1 pattern), thereby suppressing alternans.

In general, feedback control of dynamical systems involves measuring the state of the system (characterized by APD in our model), generating an appropriate feedback signal, and adjusting an actuator that modifies an accessible system parameter (PP in our model) by an amount ε_n . The size of the feedback signal can be small and little energy is expended by the controller when the system is stabilized to the desired state because the scheme stabilizes a state that already exists in the system but is unstable in the absence of control.

To illustrate suppression of alternans using our simple mathematical model, we consider closed-loop proportional feedback. Proportional control seeks to minimize the difference between the current state of the system and a reference state, which we take as APD^* . The algorithm uses an error signal given by

$$\varepsilon_n = \gamma(APD_n - APD^*), \quad (8)$$

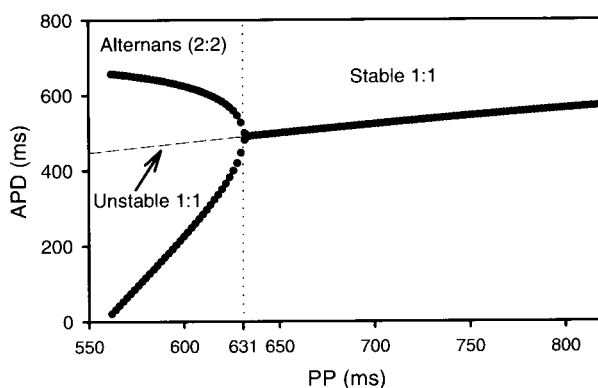


Fig. 9. A numerically generated bifurcation diagram for a simple model of periodically paced cardiac muscle. For PP larger than 631 ms, the response pattern is 1:1. A supercritical bifurcation occurs at 631 ms, leading to alternans for faster pacing. The dashed line indicates the location of the unstable 1:1 pattern.

where γ is the feedback gain, which is used to adjust in real time the pacing period at each iterate as

$$PP_n = PP^* + \varepsilon_n, \quad (9)$$

where PP^* is the nominal pacing period. Note that $\varepsilon_n = 0$ when control is successful ($APD_n = APD^*$) so that $PP_n = PP^*$. Therefore, the controller does not modify the location of the fixed point, only its stability.

The dynamics of the tissue in the presence of control is given by

$$APD_{n+1} = f(PP_n - APD_n), \quad (10)$$

whose steady-state solutions are still equal to APD^* since the controller does not effect the location of the fixed point of the system. Its behavior in a neighborhood of the fixed point is governed by

$$\delta APD_n = \mu(1 - \gamma)\delta APD_n. \quad (11)$$

To suppress alternans and stabilize the 1:1 pattern, the feedback gain must be chosen so that the condition $|\mu(1 - \gamma)| < 1$ is satisfied. When $\gamma = 1$, any perturbation from the fixed point δAPD_n is driven to zero on the very next iterate of the map. Note that the presence of chaos is not required for this method to be successful.

A simple numerical illustration of a proportional control experiment is shown in Fig. 10. The map (10) is iterated initially with $\gamma = 0$ for $PP^* = 600$ ms and APD_n is plotted as a function of n . This tissue model displays alternans with $APD^* = 473.96$ ms and $\mu = -1.084$ so that it should be possible to stabilize the 1:1 response pattern for $0.076 < \gamma < 1.92$. Control is initiated with $\gamma = 0.15$ as indicated by the vertical dashed line (Fig. 10a).

Initially, large adjustments to the pacing period are made that are designed to direct the system to the 1:1 state. Eventually, the perturbations become vanishingly small as the system approaches the fixed point. Once control is turned off (vertical dotted line), the fixed point is once again unstable and the system returns to alternans. The transient approach to the fixed point is rather long in this situation since the feedback gain is chosen close to the boundary of the domain of control. It can be made significantly shorter by choosing the gain closer to the optimum value as shown in Fig. 10b for $\gamma = 0.9$. Experiments demonstrating control of alternans in small pieces of periodically paced cardiac muscle using a modified control algorithm that adapts to changes in the tissue parameters are described in Section 7.

For simplicity, we have considered feedback control of a fixed-point of a one-dimensional dynamical system. Successful application of these control methods is not limited to such simple systems. Recent research has investigated the application of other linear and non-linear control schemes to moderate-dimension systems [34,35].

While the concepts described in this section are intriguing, additional research is needed to address how to apply these methods to the whole heart during fibrillation where spatial complexity is an issue. In addition, it is difficult to modify the local pacing period in the whole heart so different methods for modulating the local

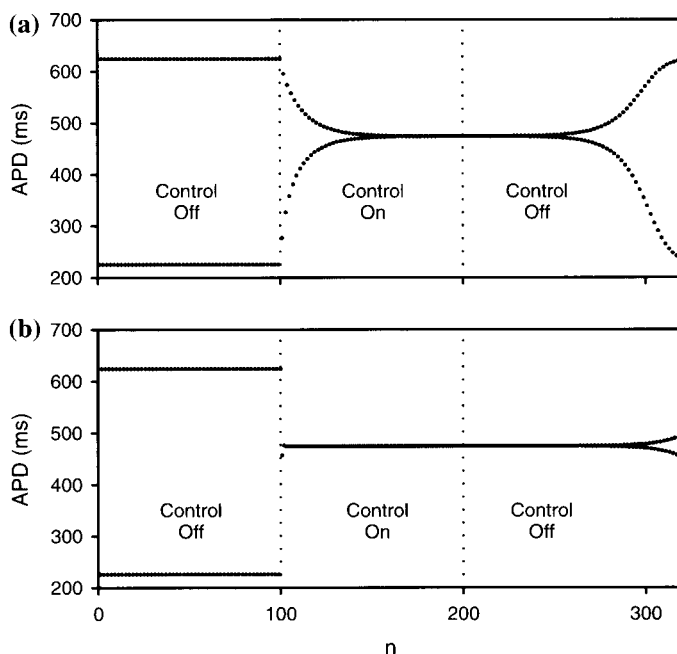


Fig. 10. Temporal evolution of APD as a function of the beat number n demonstrating the effects of feedback control. Control is turned on during the interval between the vertical dotted lines with (a) $\gamma = 0.15$ and (b) $\gamma = 0.9$.

cardiac dynamics are needed. In the next section, we review some of the current research directions.

4. Controlling cardiac dynamics

The idea that small electrical stimuli can affect the dynamics of the heart is not new. It is well known that arrhythmias such as atrial flutter and ventricular tachycardia can be initiated and terminated by one or more properly timed stimuli [36]. Unfortunately, attempts to interrupt atrial or ventricular fibrillation have been less successful. Allesie et al. [37] have successfully entrained the dynamics of a spatially localized portion of the myocardium during atrial fibrillation using rapid pacing. This procedure did not result in defibrillation, however; complex dynamics reappeared after the pacing was terminated. Similarly, KenKnight et al. [38] captured the local dynamics during ventricular fibrillation but also did not achieve defibrillation.

Approaching the problem from a different perspective, Garfinkel et al. [3] have demonstrated that it is possible to stabilize cardiac arrhythmias in an *in vitro* heavily medicated small piece of the intraventricular septum of a rabbit heart by administering small, occasional electrical stimuli. The protocol is referred to as proportional perturbation feedback (PPF) and it is a variation of the feedback scheme described

in the previous section. In terms of the behavior of the heart, the scheme uses feedback to stabilize periodic beating (the unstable dynamical state) and destabilize complex rhythms using small perturbations. Crucial to this strategy is the concept that the heart can display two different dynamical behaviors under essentially identical physiological conditions, such as normal sinus rhythm and fibrillation or tachycardia and fibrillation, for example. Note that variations of this method for controlling cardiac dynamics have been suggested [39], some of which are simpler and more robust [40]. The PPF control strategy has also been used by Schiff et al. [41] to stabilize the electrical behavior of the rat hippocampus, suggesting that it may be possible to develop an intervention protocol for epilepsy.

While these results are intriguing, they are surrounded by some controversy because of the criterion used to detect chaos in the biological preparations. Pierson and Moss [42] recently investigated the influence of noise on the analysis procedure and found that it is indeed capable of detecting unstable periodic orbits even in the presence of large amounts of noise. Also, Christini and Collins [43] have suggested that PPF can be used even in situations where the dynamics is driven by stochastic, rather than deterministic, influences; they require only the existence of unstable periodic orbits.

More recently, Hall et al. [4] demonstrated that an adaptive controller can be used to suppress temporal instabilities in an atrial-ventricular nodal conduction system known to exhibit alternans. The control protocol is based on a comparison of the most recently observed interbeat interval with the previous interval. This method has been analyzed by Gauthier and Socolar [44] and is similar to the one used to control the dynamics of physical systems [31]. Note that the results of Garfinkel et al. [3] and Hall et al. [4] only address specialized systems of the heart but not necessarily the muscle of atrial or ventricular walls, which are the primary substrates for fibrillation.

While these results are intriguing, they demonstrate only that *temporal* complexity of a dynamical system can be controlled. There is not yet a general approach to controlling systems that display *spatio-temporal* complexity such as that displayed by the heart during fibrillation [8,9,10]. Toward this goal, Glass and Josephson [45] established criteria for resetting and annihilation of reentrant arrhythmias; Biktashev and Holden [46] demonstrated that proportional feedback control can induce drift of spiral waves in a model of cardiac muscle; Aranson et al. [47] showed that external stimuli can stabilize meandering spiral waves; Watanabe and Gilmour [39] proposed a strategy that uses small stimuli to prevent cardiac rhythm disturbances, and Fenton et al. [12] showed that breakup of spiral wave patterns can be suppressed by applying electrical stimuli at many spatial locations. In addition, preliminary research by Ditto and collaborators [13] suggests that it is possible to capture at least a portion of a fibrillating human atrium using the PPF method, where the heart dynamics is recorded and control stimuli are delivered through a quadrupolar electrode catheter inserted into the right atrium through the femoral vein. Due to the limited number of sensors, it could not be determined whether the entire atrium was controlled. Unfortunately, atrial fibrillation was terminated only rarely using this method.

There are several issues that must be addressed before these recent results can be put to practice. In the work of Garfinkel et al., Biktashev and Holden, Watanabe and Gilmour, and Fenton et al. the use of spatially uniform forcing function or application of perturbations at numerous spatial locations is assumed, a requirement that is unlikely to be satisfied in practice. Also, the results of Aranson et al. and Glass and Josephson were obtained using a simplified model of cardiac dynamics whose properties are, in many aspects, different from the properties of cardiac muscle. Finally, experiments using many sensors at different spatial locations [9,10,24–27] are needed to determine whether the methods used in the preliminary work of Ditto and collaborators can successfully capture the entire atrium.

Our approach to addressing these issues involves several steps. We believe that fibrillation can be controlled optimally in a clinical setting using feedback control methods at one or a few spatial locations on or in the heart, or by a combination of control and synchronization methods. To achieve this long-term goal, we believe that it is crucial to develop a quantitative understanding of how control stimuli interact with cardiac muscle that has passed through a bifurcation since these bifurcations are thought to be responsible for initiating fibrillation [48]. Our first step is to study small pieces of cardiac muscle where spatial complexity is not important. In these studies, precise experiments will be compared quantitatively to ionic-based cardiac mathematical models. The next steps involve extending the work to one, two, and three spatial dimensions, both experimentally and theoretically. The following sections describe our ongoing research investigating the dynamics and control of small pieces of cardiac muscle.

This methodical approach of comparing experiments and mathematical models starting with 'simple' preparations has been followed by several research groups with great success. For example, phase resetting and phase locking in rapidly paced chick cell aggregates can be predicted accurately by an ionic model as demonstrated by Clay et al. [49] and Kowtha et al. [50], respectively; Wenckebach periodicities in rapidly paced enzymatically dissociated guinea pig ventricular myocytes can be predicted by the Beeler–Reuter model as demonstrated by Delmar et al. [51]; Roth [52] has shown that the induction of reentry by fast pacing through unipolar electrodes observed by Lin et al. [53] can be explained using a two-dimensional bidomain model with unequal anisotropy ratios and the Beeler–Reuter membrane dynamics; and Roth and Krassowska [54] have obtained quantitative agreement between experiment and theory for the induction of reentry in cardiac tissue by taking into account the manner in which electric fields alter the transmembrane potential. We believe that this approach is also important for research on controlling cardiac dynamics using small perturbations, consistent with the advice of Holmes during his presentation at the Workshop on Mathematical Approaches to Cardiac Arrhythmias,

'Many of these [nonlinear dynamics] methods are in their infancy and we are unaware of their limitations. It is therefore best to proceed with caution and to attempt careful studies of 'simple' systems under well-controlled conditions (unless we are more interested in press coverage than scientific progress)' [55].

5. Prevalence of rate-dependent behaviors

We have investigated the dynamics of small pieces of periodically paced cardiac muscle without the possibility of spatial instabilities before investigating how the spatially localized behaviors fit into the more complex spatially extended behaviors. We measure the tissue response while varying parameters such as the pacing interval and strength in a systematic way. For example, under various conditions, S stimuli can elicit R responses (S:R behavior). A wide range of these responses have been observed depending on the type of cardiac tissue, animal species, or stimulus parameters, such as frequency, amplitude, and shape.

In order to determine the prevalence of different rate-dependent behaviors, we investigated the response of small pieces of bullfrog (*Rana catesbeiana*) ventricular myocardium to periodic electrical stimulation [22]. We concentrated on *rate-dependent* behaviors in cardiac muscle because of the wide range of excitation rates that occur in both healthy and pathological cardiac tissue. We explored the range of dynamical behaviors in a large number of animals to determine the relative prevalence of different dynamical states because control protocols must be able to deal with every behavior occurring in a population.

In our experiments, the heart was excised from adult animals of either sex and from 4 to 8 in. in length². After pacemaker cells were cut away, a small piece ($\leq 3 \times 3 \times 5$ mm) of ventricular myocardium was removed, placed in a chamber, and superfused with a recirculated physiological solution. The tissue, quiescent in absence of an applied stimulus, was paced with 4 ms-long square-shaped current pulses applied through two fine (51 μ m) tungsten wires set ~ 2 mm apart on the surface of the tissue. The amplitude of the applied stimulus was typically ~ 0.1 – 0.2 mA, twice the current necessary to elicit a response so that the experiments were conducted in a parameter regime where only 1:1, 2:2 and 2:1 dynamics should occur [56]. Intracellular (transmembrane) voltages were typically measured within 1–2 mm of the stimulus electrodes using a glass micropipette. Before initiating data collection, the tissue was paced at PP of ~ 1000 ms for about 20 min.

A bifurcation diagram is a convenient way to summarize observations of the long-term dynamical behavior of the tissue as PP is varied slowly. We obtained such a diagram by recording APD while adjusting PP across a wide range of physiological values, from 1200 to 300 ms in 100 or 50 ms intervals (downsweep), and then from 300 to 1200 ms, again in 100 or 50 ms steps (upsweep). For each PP, the response of the tissue to the first 5–10 stimuli was discarded in order to eliminate

2 All procedures are approved by the Duke University Institutional Animal Care and Use Committee (IACUC) and conform to the Research Animal Use Guidelines of the American Heart Association. Bullfrogs are anesthetized using 1% wt/vol 3-aminobenzoic acid ethyl ester mixed with cold tap water, double pithed, and dissected by cutting along the ventral side of the body. Physiological solution (100 ml/min flow rate) is maintained at temperature $20 \pm 2^\circ\text{C}$ and contains 110 mM NaCl, 2.7 mM KCl, 1.5 mM MgCl_2 , 1.8 mM CaCl_2 , 5.6 mM glucose, 2.8 mM Na_2HPO_4 and 1 mM HEPES. It is buffered with NaHCO_3 and bubbled continuously with 95% O_2 and 5% CO_2 to maintain a pH of 7.4 ± 0.1 . To reduce motion of the tissue due to contractions, 5–20 mM 2-3-butanedione monoxime (DAM) is sometimes added. We find that DAM does not affect the gross dynamical features of this preparation.

transients and the subsequent behavior was recorded for up to 10 s. After discarding transients, the width of each action potential was determined at 70% of full repolarization and plotted at each PP.

Fig. 11a shows a bifurcation diagram for an animal that did not display alternans (2:2 pattern). For long PP, the tissue responds in a 1:1 pattern and it is seen that the APD decreases for decreasing PP (open circles). As the PP was decreased from 500 to 400 ms, an abrupt increase in the APD occurred, highlighted by the thin vertical arrow, indicating the bifurcation from a 1:1 to a 2:1 pattern. After the smallest PP of our pacing protocol was achieved, the PP was increased slowly (closed triangles). The tissue remained on the 2:1 branch for a wide range of PP values which elicited 1:1 response during the downsweep. At a PP between 700 and 800 ms, the tissue response makes an abrupt transition back to 1:1 behavior. The coexistence of two stable behaviors (response patterns) for a single system parameter (the PP in this case) is known as *bistability*.

We found that bistability between 1:1 and 2:1 patterns exists in 17 of 23, or 74%, of the cardiac preparations studied. The window of bistability was situated over a range of PPs that lies near the natural period of the resting heart (~ 1000 ms), and extends for ~ 160 ms. We found that 2:2 behavior (alternans) occurs in 8 of 23 or 35% of cardiac preparations. In all cases where 2:2 behavior was observed, bistability also occurs between 2:2 and 2:1 dynamics. We found that the stimulus that does not elicit a response in the 2:1 state does, in fact, have an effect on the tissue dynamics. Furthermore, we found that the bifurcation to 2:1 behavior is subcritical, and that unstable periodic orbits did not occur for pacing intervals shorter than that at which the tissue undergoes a transition to a 2:1 pattern. Most significantly, our findings indicate that bistability is highly prevalent in bullfrog cardiac tissue, and thus optimal controllers must be designed with sufficient flexibility to deal with bistability as well as unstable periodic orbits.

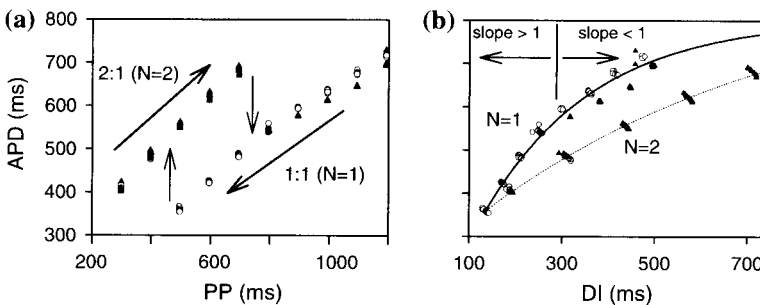


Fig. 11. (a) Bifurcation diagram showing 2:1 $\longleftrightarrow$ 1:1 bistability and hysteresis. (b) Plot of APD as a function of DI for the data shown in (a). The vertical line at $DI = 285$ ms indicates the diastolic interval at which the slope of $N = 1$ branch becomes greater than unity. For both plots, open circles (closed triangles) are APD measurements as the PP is decremented (incremented).

We attempted to fit our observations to the simple one-dimensional mapping (3). Note that the 1:1 and 2:2 patterns observed in the experiments correspond to $N=1$, and the 2:1 pattern corresponds to $N=2$. This model assumes that a stimulus that fails to elicit an action potential has no effect on the subsequent dynamics. The mapping (3) predicts that the 1:1 behavior ($N=1$) should be unstable when $|\mu| > 1$ and that, in a plot of APD as a function of DI, all data points should fall on a single curve (defining f).

As shown in Fig. 11b, which shows the dependence of f on DI_n for the data displayed in Fig. 11a, it is seen that one value of DI can give rise to two different values of APD, depending on whether the tissue response is on the $N=1$ (solid line) or $N=2$ (dotted line) branch. Note that the APD is *shorter* for the $N=2$ branch than the $N=1$ branch for the same value of DI, which we attribute to the effect of the stimuli that does not elicit an action potential. That is, this 'extra' stimulus has a rather significant effect on the subsequent action potential and shortens its duration. Hence, it should be possible to control the dynamics of the tissue by applying stimuli during the period when the tissue is refractory. In Section 6, we show that it is also possible to lengthen the APD by applying the stimulus at a different time in the refractory period. Modulating the APD by stimuli applied during the refractory period may prove to be one cornerstone of methods for controlling whole-heart dynamics.

In addition to the multi-valued nature of the tissue response, we found that the 1:1 behavior ($N=1$) was stable even when the slope of the APD vs. DI curve was greater than one, as indicated in Fig. 11b. We found that a difference-differential model proposed by Chialvo et al. [57], incorporating a 'memory effect,' could explain this behavior, but it did not correctly predict the transition from the $N=2$ to the $N=1$ branch.

6. Inducing transitions between bistable states

Since modulation of complex local dynamics in the heart may be critical for suppressing the initiation of or controlling the spatio-temporal complexity of fibrillation, we investigated methods of manipulating experimentally the bistable 1:1 and 2:1 states [58]. In much earlier work, Mines had shown that it was possible to cause a 2:1 $\rightarrow$ 1:1 transition with a single stimulus injected into the train of applied pulses [59], and Guevara et al. [60] have cited the experimental observation of 2:1 $\rightarrow$ 1:1 and 1:1 $\rightarrow$ 2:1 transitions by the injection of single stimuli into an applied train. We have extended the work of Mines and Guevara et al. by determining the range of timings of added stimuli capable of eliciting such transitions.

We found that 2:1 $\rightarrow$ 1:1 and 1:1 $\rightarrow$ 2:1 transitions can be induced repeatedly by stimuli applied over a wide range of timing intervals. The extra stimuli that induce transitions between bistable states are of the same amplitude and duration as the regular pacing applied to the tissue: only their timing is altered within the periodic pacing. This can be considered a phase-sensitive open-loop controller because inducing a transition between states depends on the relative phase at which the extra stimulus is applied.

In order to achieve transitions between the 1:1 and 2:1 states, we first determined the range of pacing intervals that yield both 1:1 or 2:1 behavior, and then paced the tissue at a PP within this window. The tissue can be biased onto the 1:1 or 2:1 branch within the bistable regime by sweeping the PP down and up until the tissue responds with the desired behavior. Once the tissue was in the 1:1 or 2:1 state, we injected an ' S_2 ' stimulus at an interval T after the preceding normal stimulus. The next normal stimulus was applied at an interval $PP - T$ following the S_2 pulse, marking the resumption of the normal stimulus train. We observe whether or not a transition takes place. This procedure is repeated for various PP values and T values in order to map the domain in the (PP, T) plane over which transitions can be induced. The interval T is varied from 50 ms to $PP - 50$ ms in 50 ms increments.

Fig. 12 shows one example of a 1:1 $\rightarrow$ 2:1 transition where the S_2 stimulus (marked with an arrow) is applied during the early part of an action potential. This *lengthens* the action potential's refractory period long enough to cause the next regular stimulus to be ineffective, giving the tissue a time interval equal to $2PP - T$ over which to recover, extending the next action potential after which it falls into a 2:1 pattern. This observation demonstrates that stimuli added during the tissue's refractory period can have a dramatic effect on the subsequent dynamics.

We observed that there are two different types of 1:1 $\rightarrow$ 2:1 transitions as well as two distinct types of 2:1 $\rightarrow$ 1:1 transitions, each distinguished by their transient behavior en route to steady-state dynamics. A qualitative understanding of the nature of these transitions can be motivated using the simple one-dimensional iterative mapping model described in Section 3, which gives good qualitative predictions for three of the four transition types we observe.

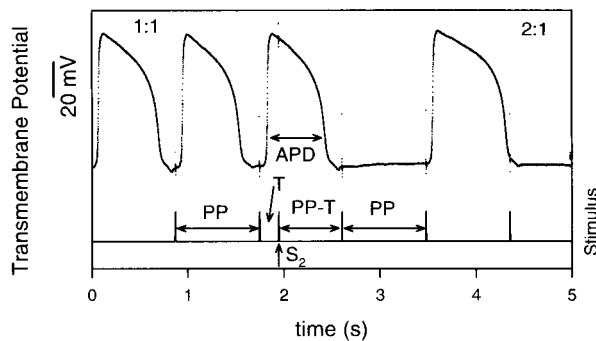


Fig. 12. Temporal evolution of the transmembrane potential showing a 1:1 $\rightarrow$ 2:1 transition. The stimulus waveform is shown at the bottom of the figure and the extra stimulus is marked with an arrow.

7. Controlling alternans

Using the feedback control methods discussed in Section 4 and our experimental test-bed, we have demonstrated that it is possible to stabilize an unstable 1:1 response pattern for a PP that produces alternans (2:2 pattern) when the control is turned off. The 1:1 state was stabilized using two variations of the feedback scheme known as time delay auto-synchronization (TDAS) [31]. This protocol has a short memory that allows it to automatically track slow changes in the stability of the dynamics. Note that Hall et al. [4] used a similar scheme for stabilizing alternans in AV-nodal conduction times, and Socolar and Gauthier [61] have analyzed theoretically by an extension of the protocol.

Feedback protocols attempt to stabilize an unstable state of the system by making small adjustments to an accessible system parameter based on an estimate of the distance from the current state of the system to a desired state. In our experiment, the state of muscle tissue was characterized by measuring the n -th action potential duration (APD_n), and the n -th pacing period (P_n) was used as the accessible system parameter. An error signal ε_n is generated using the TDAS algorithm according to

$$\varepsilon_n = \gamma(APD_n - APD_{n-1}), \quad (12)$$

where the past behavior of the system is used to estimate the location of unstable state to be stabilized. The n th pacing period PP_n is perturbed around its nominal value PP^* as

$$PP_n = PP^* + \varepsilon_{n-1}. \quad (13)$$

When control is successful, $APD_n = APD_{n-1} = APD^*$, which is the desired 1:1 behavior, and $\varepsilon_n \rightarrow 0$. Hence, the controller does not effect the state of the system, only its stability.

Fig. 13 shows a complete TDAS control experiment. The temporal evolution of a monophasic action potential recorded from the tissue and the error signal are shown for a PP that produces alternans as control is turned on and off, indicated by the vertical dashed lines. With control off (to the left of the first dashed line) the APD_n alternates between approximately 110 and 180 ms. When the feedback perturbations are applied to the tissue (in between the dashed lines), the response pattern converges to 1:1 behavior with $APD^* = 132$ ms after a few second transient and ε_n attains a small value ($<0.002 PP^*$). When control is turned off (to the right of the second dashed lines), the 1:1 dynamical state is unstable and the system slowly returns to alternans. This behavior is reminiscent of the numerical experiments shown in Fig. 10.

In contrast to PPF algorithms [3] that uses a comparison between the current state of the system and a reference state (the desired unstable state) determined through a time-consuming 'learning phase,' TDAS has the advantage of using feedback generated solely from a comparison between current and previous states of the system. Thus TDAS algorithms, which are less sensitive to parameter drift and

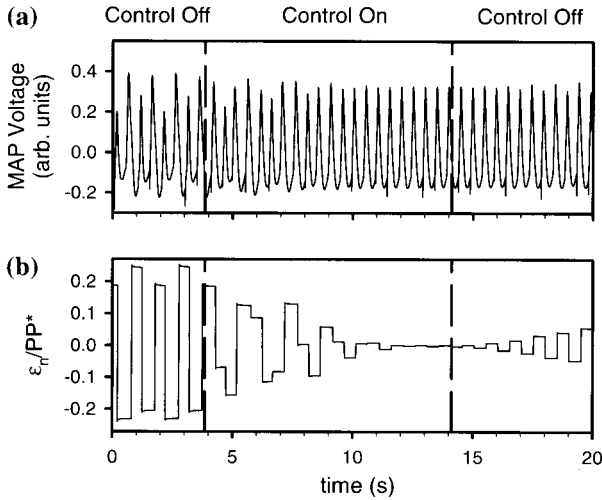


Fig. 13. Temporal evolution of the (a) monophasic action potential signal (MAP) and (b) the error signal ε_n (quantified as a fraction of the average period PP^*) during a control experiment.

experimental noise than PPF methods, have potential advantages for the control of complex behavior in biological systems.

8. Outlook

Our preliminary experiments on controlling alternans in the bullfrog tissue is a first step toward realizing a method for controlling complex whole-heart dynamics. We are currently investigating the use of variants of this method to control the dynamics of a fibrillating sheep atrium in conjunction with a high-density mapping system to record the effects of the control perturbations on the entire atrium. This basic research program will provide new insights into the fundamental mechanisms that regulate the dynamics of small pieces of cardiac muscle subjected to small, appropriately timed electrical stimuli. The studies will help to elucidate the important factors that determine the success or failure of nonlinear dynamics control techniques in stabilizing cardiac arrhythmias. The knowledge gained from these studies may improve current clinical methods of cardioversion and will increase the number of treatment options for cardiac patients.

Acknowledgements

We gratefully acknowledge invaluable collaborations with Ellen Dixon-Tulloch, Wanda Krassowska, Robert Oliver, and Patrick Wolf of the Duke Experimental Electrophysiology Laboratory, Henry Greenside and Joshua Socolar of the Duke

Physics Department, and David Schaeffer of the Duke Mathematics Department, and financial support of the Whitaker Foundation and the National Science Foundation.

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